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Functional Requirements of Modules

User Management

This module allows the management of three types of users. These are: admin, customer and guest. Customers and guests have similar functionalities only that guest have lesser than the former.

- Allow users to log in
- Allow users to sign up
- Verify the credentials of user
- Check for repeating usernames while creating an account
- A cart should be created for customer each time a sign up is made
- Guests can access the program without a need to enter any details
- While browsing the cart, users should be able to filter out the items so they can easily select the item they wish to add to their carts. Such as filter by category
- Users should be allowed to reset their passwords
- While editing their profile, users should verify their accounts for security purposes

History Module

This module allows the customers to view what items they have bought before. It provides a brief yet valuable overview of their past purchases. With the help of this module, customers can track what items they have bought previously

- Customers can access history module from their homepage
- Guests can not access history module
- View history should show all the items previously bought by the customer
- Customers can willingly clear all history
- The item should become available in the history module as soon as the customer checks out the cart.
- History should not be allowed to be changed by customer
- Customers can only view their own purchase history and not of other customers'
- For privacy measures, admin also can not access the purchase history of a customer

- Customers should be displayed the cart number of each item so customer knows what items were ordered with what other items in the previous access to cafe

Items Module

This module contains all the data of items available in cafe. It manages them through a database with multiple tables. Items module allows customers/guests to view menu of cafe and allows the admin to edit it.

- Customer and guests should be allowed to browse items
- Only user who can edit add remove and update the items module should be admin
- Provide admins with the ability to set multiple pricing options for items, such as regular price and discounted price to accommodate different customer segments and promotions.
- Users should be able to sort the items by their name, price, quantity etc.
- When a customer buys a product, the quantity of that product should be reduced as much as customer has bought
- Allow users to adjust item quantities directly from the cart page, providing flexibility and convenience in managing their orders.
- Implement functionality to apply promotional discounts or coupons to cart items, allowing customers to redeem special offers during checkout.
- Each product in module should belong to a category
- The UI should be pleasing and kept simple for ease of access

Cart Module

This module allows customers and guests to store items in cart. They can then proceed the cart to checkout and place the order.

- Cart should be allotted to the customer as soon as he signs up or as soon as he checks out another cart
- A new cart should be allotted to a guest each time he accesses the application
- Guests and Customers should be allowed to add, edit, remove items from the cart
- When user adds an item that is already added in the cart, the quantity of that item should increment instead of adding a new entry

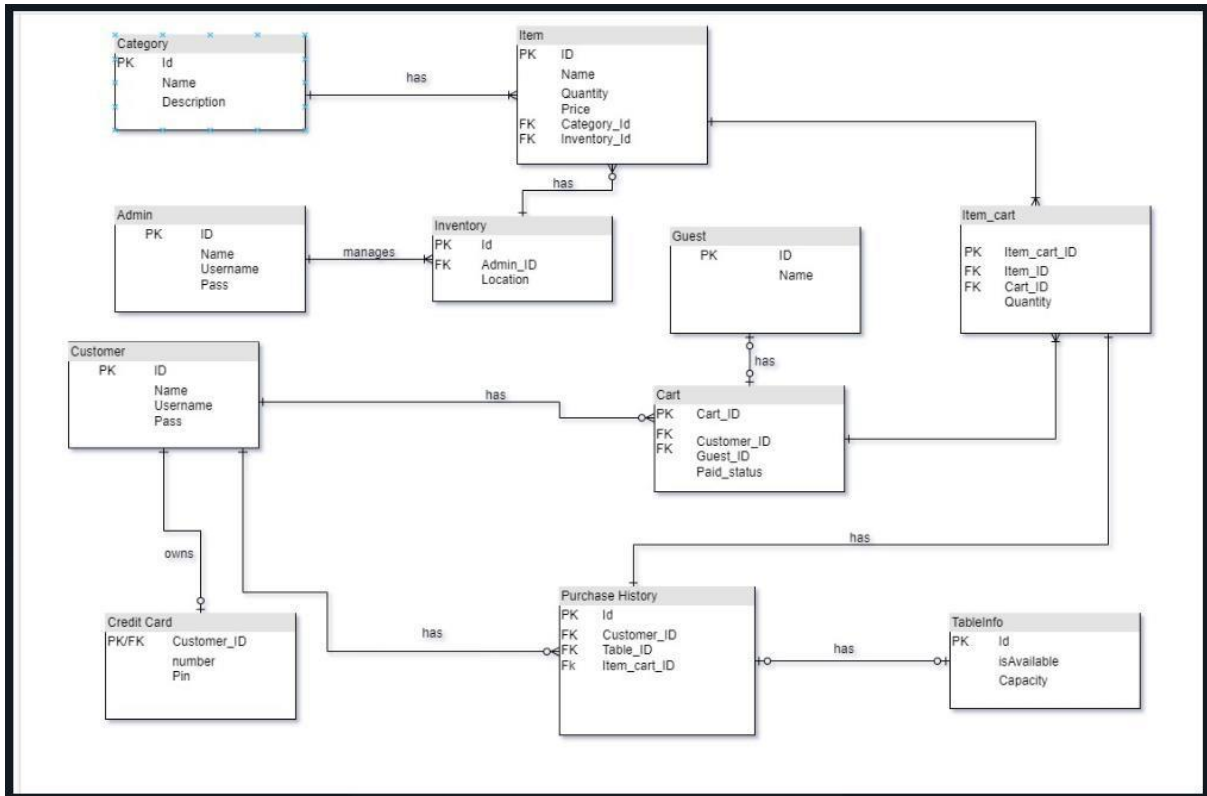
- Users should have an option to view their cart and view how much it costs right before the checkout option

Error Handling and Logging Module

This module helps in the ease of use of application. Upon wrong workings or bugs, the application gives error to the customer and displays message box. Otherwise to show the working of another module, message box is again displayed to verify its working.

- Implement comprehensive error handling mechanisms to gracefully handle unexpected errors
- Log system activities, errors, and exceptions for troubleshooting and auditing purposes
- Ensure that sensitive information is not logged to maintain data privacy and security
- Ensure that upon logging out of the system or an incorrect credentials, the input is emptied
- Ensure that user is displayed valuable messages when accessing the application (e.g. placing order)

Entity Relation Diagram



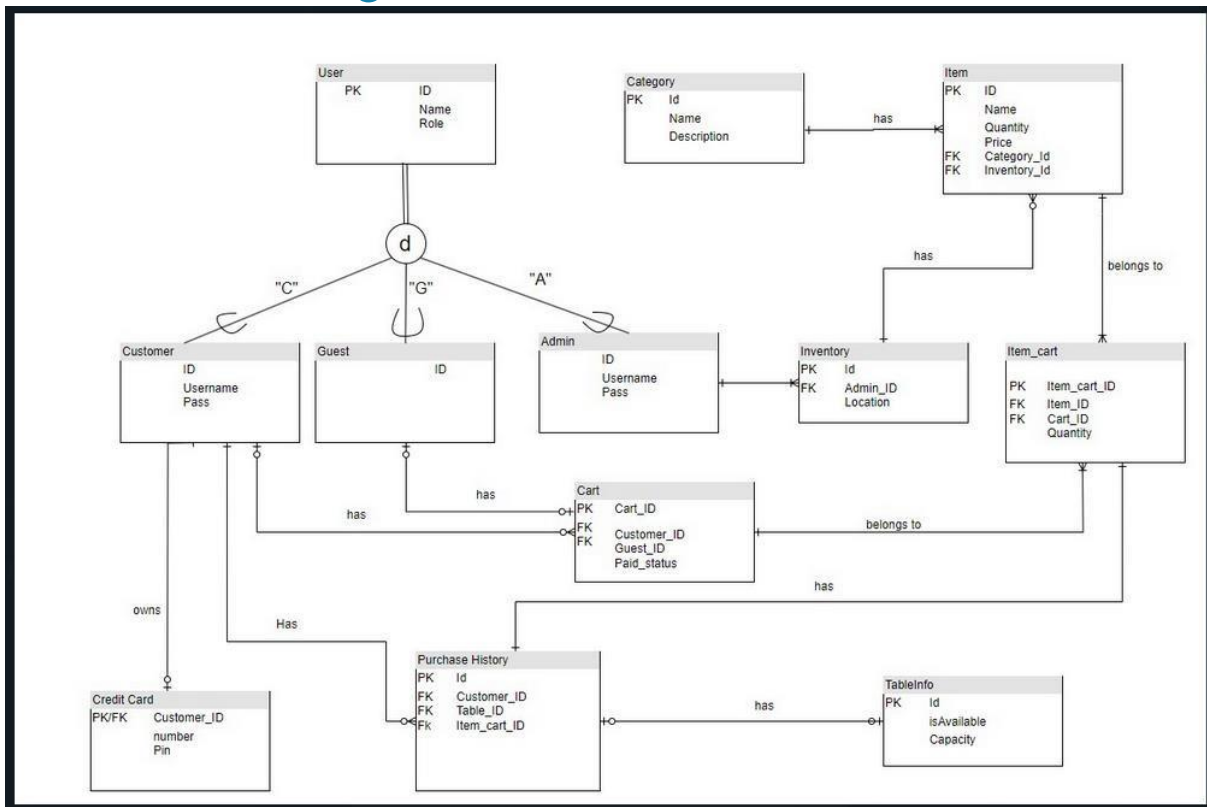
Made with draw.io

ER diagram includes following tables:

- **userCustomer**: Stores information about registered customers, including their name, username, password, and whether they have a card associated with their account.
- **userAdmin**: Stores information about administrators, including their name, username, and password.
- **userGuest**: Records information about guests, including their name.
- **Category**: Contains categories for items available in the cafe, with a unique description for each category.
- **inventory**: Tracks inventory details, such as location, associated administrator, and inventory ID.
- **item**: Stores information about cafe items, including name, quantity, price, associated category, and inventory ID.
- **Cart**: Manages shopping carts, including customer or guest ID, payment status, and whether the cart is paid.
- **TableInfo**: Records information about tables in the cafe, including capacity, availability status, and the associated customer ID if occupied.

- **item_cart:** Maps items to specific shopping carts, including item ID, cart ID, quantity, and item name.
- **purchaseHistory:** Tracks the purchase history of customers, including the item_cart ID, customer ID, and table ID for each purchase.
- **CreditCard:** Stores credit card information for customers, including customer ID, PIN, card number, and expiry date.

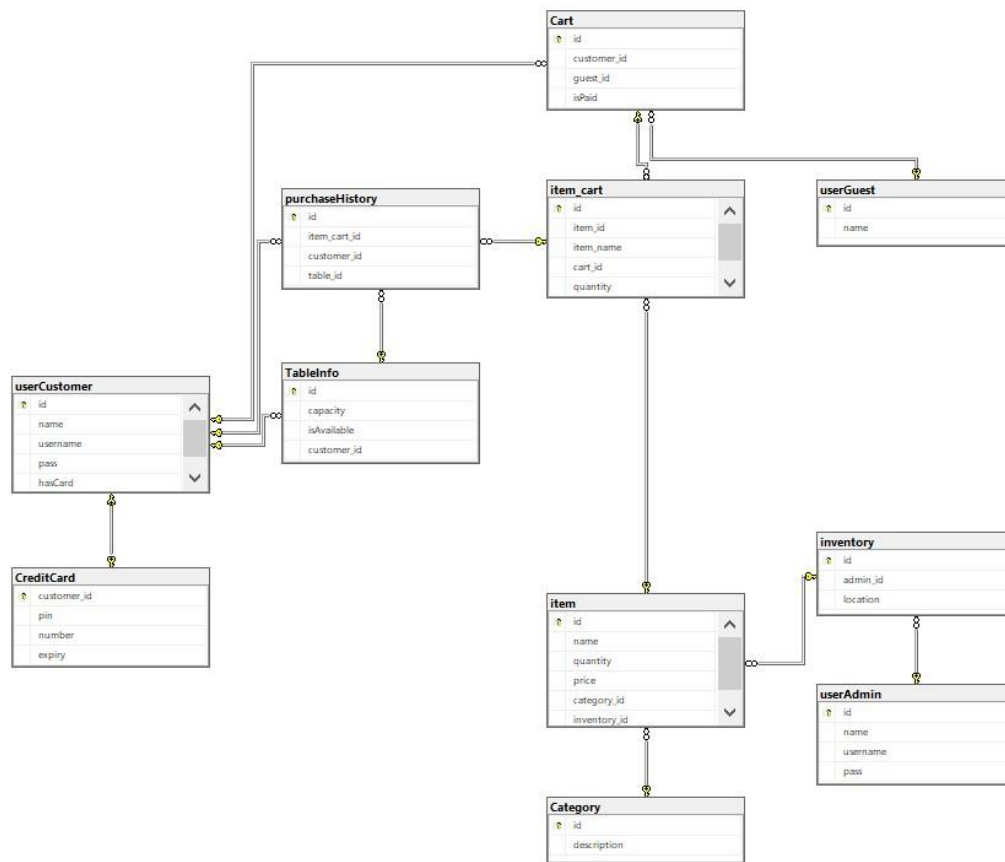
Enhanced ER Diagram



Made with draw.io

The EERD includes the generalization of users into one user entity. The types of user include userCustomer, userGuest, userAdmin. Customer and guest have similar functionalities and accessibilities but the former entity having access to more functions. Meanwhile, the admin manages the inventory and can perform other functions related to the cafe.

Relational Schema



Made in ssms **userCustomer**:

Attributes: (id(pk), name, username, pass, hasCard)

Related Table: **Cart** (customer_id)

Related Table: **purchaseHistory** (customer_id)

Related Table: **CreditCard** (customer_id)

userAdmin:

Attributes: (id(pk), name, username, pass)

Related Table: **inventory** (admin_id)

userGuest:

Attributes: (id(pk), name)

Related Table: **Cart** (guest_id)

Category:

Attributes: (id(pk), description)

Related Table: item (category_id)

inventory:

Attributes: (id(pk), admin_id, location)

Related Table: item (inventory_id)

item:

Attributes: (id(pk), name, quantity, price, category_id, inventory_id)

Related Table: item_cart (item_id)

Cart:

Attributes: (id(pk), customer_id, guest_id, isPaid)

Related Table: item_cart (cart_id) TableInfo:

Attributes: (id(pk), capacity, isAvailable,

customer_id) Related Table: purchaseHistory

(table_id) item_cart:

Attributes: (id(pk), item_id, item_name, cart_id, quantity)

Related Table: purchaseHistory (item_cart_id)

purchaseHistory:

Attributes: (id(pk), item_cart_id, customer_id, table_id) CreditCard:

Attributes: (customer_id(pk), pin, number, expiry)

Related Table: userCustomer(customer_id)

Normalization

First Normal Form (1NF):

In the given schema:

- All tables have a well-defined primary key.
- Each attribute within the tables contains atomic (indivisible) values.
- There are no repeating groups within the tables.

- Each column contains only one value per record.
- We have considered Name property as atomic.

Second Normal Form (2NF):

In the given schema:

- All tables are in 1NF.
- Non-key attributes are fully dependent on the entire primary key.
- There are no partial dependencies.
- For example, in the **userCustomer** table, attributes like **name**, **username**, **pass**, and **hasCard** are fully dependent on the primary key **id**.

Third Normal Form (3NF):

In the given schema:

- All tables are in 2NF.
- There are no transitive dependencies between non-key attributes.
- For example, in the **userCustomer** table, there are no transitive dependencies between **id**, **name**, **username**, **pass**, and **hasCard**. Each non-key attribute is directly dependent on the primary key **id**, and there are no dependencies between non-key attributes.

Documentation and User Guide

Introduction

The Fast Cafe Management System is a comprehensive solution designed to streamline operations and enhance the customer experience in cafes. It offers three main dashboards tailored to different user roles: Customer, Admin, and Guest.

Admin Dashboard

1. Menu Management:

- Add, edit, or remove items from the menu.
- Categorize items for easy navigation.
- Update item details such as name, description, price, and availability.

2. Category Management:

- Add new categories to the menu for better organization.

- Remove categories that are no longer needed.
- Edit category names or descriptions as required.

3. Discount Management:

- Apply discounts on entire categories to boost sales.
- Set discount percentages for specific categories.
- Remove discounts as needed.

4. Table Management:

- Manage table availability and capacity.
- Mark tables as available or reserved.
- Update table capacities as needed for efficient seating.

Customer Dashboard

1. Menu Navigation:

- Upon logging in, you will be directed to the Customer Dashboard.
- Browse through the menu categories by selecting from drop down such as Coffee, Tea, Pastry, Sandwiches, Smoothies, and Salads.

2. Placing an Order:

- Click on the item you wish to order.
- Specify the quantity and any customization options (if available).
- Click the "Add to Cart" button.
- Review your order in the cart and click "Proceed to Checkout".

3. Checkout Process:

- Review your order details and make any necessary changes.
- Choose your payment method (Credit Card or Cash).
- If paying by credit card, enter your payment details securely.
- Click "Place Order" to confirm your purchase.

4. Account Management:

- Update your account information, such password, under "Account Settings".
- View your order history to track previous purchases.
- Add card information securely for faster checkout.

Guest Dashboard

1. Menu Navigation:

- Browse the menu categories and select items you wish to order.

2. Adding to Cart:

- Click on the item, specify quantity, and click "Add to Cart".

3. Checkout Process:

- To proceed, log in to your existing account or sign up.
- Review your order and choose the payment method.
- Complete the checkout process.

4. New User:

- If new, sign up to create an account.
- Fill out the required information and proceed to checkout.

Screenshot of Implementation

Admin Dashboard Implementation

[Signu](#) [Continue as a Guest](#)

Welcome to FAST's Cafe System

Username

Password

☐ Customer ☒ Admin

Login

Welcome *admin*

	ItemID	ItemName	Quantity	Price	Category
▶	1	Espresso	20	2.50	Coffee
	2	Cappuccino	30	3.00	Coffee
	4	Americano	100	3.00	Coffee
	5	Mocha	20	10.00	Coffee
	6	Green Tea	50	2.00	Tea

	id	name	quantity
▶	1	Espresso	20
	2	Cappuccino	30
	4	Americano	100
	5	Mocha	20
	6	Green Tea	50
	7	Black Tea	60
	9	Chai Tea	50
	10	Herbal Tea	90
	11	Croissant	100
	12	Muffin	120
	13	Danish	300

Add Item

ID

Name:

Quantity:

Price:

Category

Add

Add Discount

Discount Percentage

Category

Add

Change Password

UserName

Old Password

New Password

Confirm New

Change

Customer Dashboard Implementation

Welcome, hasun

Browse Items

Manage Cart

View History

Edit Account

Log Out

Proceed to Checkout

Browse Items

	id	name	quantity	price
▶	1	Espresso	20	2.50
	2	Cappuccino	30	3.00
	4	Americano	100	3.00
	5	Mocha	20	10.00

Your Cart

	item_name	quantity	item_id
▶	Espresso	2	1
	Americano	6	4

Filter

Add

Manage Your Cart

	item_name	quantity	item_id
▶	Espresso	2	1
	Americano	6	4

Set Quantity

Set

Delete

Return

	customer_id	cart_id	name	quantity
▶	1	22	Latte	1
	1	23	Latte	3
•				

- Change Password
- Add Credit Card
- Clear History

Go Back

Cart

	item_id	item_name	quantity
▶	16	Turkey Sandwich	1

Total Price: \$

6.50

Payment Method

☐ Cash

☐ Credit Card

Go back

Proceed

Guest Dashboard Implementation

Welcome, GUEST

Browse Items

Manage Cart

Log Out

Proceed to Checkout

[Back](#)

Name

Username

Password

Confirm Password

Signup